

03_snapshot_alignment

```
source("R/00_setup.R")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
model_coords <- readRDS("model_coords.rds")
coords       <- readRDS("country_wave_coords.rds")

gt <- coords %>% transmute(country = iso, wave, gt1 = Dim1, gt2 = Dim2)

#' ## The matched backbone
#'
#' For every country-wave, the model's C2 coordinate joined to the true
#' coordinate for that same wave, with signed per-axis gaps and Euclidean
#' distance. Every later change analysis reads from this table, because C2 is the
#' only condition matched to the ground truth on both country and year.

c2_matched <- model_coords %>%
  filter(condition == "C2") %>%
  inner_join(gt, by = c("country", "wave")) %>%
  mutate(d1 = Dim1 - gt1,      # signed gap, tradition-secular
         d2 = Dim2 - gt2,      # signed gap, survival-self-expression)
```

```

    dist = sqrt(d1^2 + d2^2)) %>%
  select(model, country, wave, Dim1, Dim2, gt1, gt2, d1, d2, dist)
saveRDS(c2_matched, "c2_matched.rds")

c2_latest <- latest_wave(c2_matched)

#' C1 has no wave dimension - the model was asked about the country with no year
#' - so it is compared against each country's most recent truth.

latest_gt <- gt %>% group_by(country) %>% filter(wave == max(wave)) %>% ungroup()

c1_latest <- model_coords %>%
  filter(condition == "C1") %>%
  select(model, country, Dim1, Dim2) %>%
  inner_join(latest_gt, by = "country") %>%
  mutate(d1 = Dim1 - gt1, d2 = Dim2 - gt2, dist = sqrt(d1^2 + d2^2))

#' ## Point estimates
#'
#' Two features of the coordinate system set the scale for every distance below:
#' country coordinates fall roughly within  $\pm 0.7$  standardized units on each axis,
#' and the average country sits 0.40 units from the global centroid.

summarise_level <- function(df, label) {
  cat(sprintf("\n=== %s: distance to most-recent snapshot ===\n", label))
  df %>% group_by(model) %>%
    summarise(n = n(), mean_dist = mean(dist), median_dist = median(dist),
              bias_Dim1 = mean(d1), bias_Dim2 = mean(d2), .groups = "drop") %>%
    mutate(across(where(is.numeric), ~ round(.x, 3))) %>% print()
}

summarise_level(c1_latest, "C1 (country only, no year)")

```

=== C1 (country only, no year): distance to most-recent snapshot ===

A tibble: 4 x 6

	model	n	mean_dist	median_dist	bias_Dim1	bias_Dim2
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Qwen/Qwen3.6-Plus	40	0.238	0.2	0.067	0.118
2	claude-opus-4-8	40	0.177	0.156	-0.049	0.043
3	gemini-3.6-flash	40	0.153	0.138	-0.047	0
4	gpt-5.5	40	0.193	0.172	-0.046	0.085

```
summarise_level(c2_latest, "C2 (country and fieldwork year)")
```

```
=== C2 (country and fieldwork year): distance to most-recent snapshot ===
```

```
# A tibble: 4 x 6
```

	model	n	mean_dist	median_dist	bias_Dim1	bias_Dim2
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Qwen/Qwen3.6-Plus	40	0.231	0.194	0.058	0.087
2	claude-opus-4-8	40	0.182	0.161	-0.04	0.052
3	gemini-3.6-flash	40	0.151	0.125	-0.067	-0.013
4	gpt-5.5	40	0.184	0.166	-0.047	0.073

```
cat("\naverage country distance from the global centroid:\n")
```

```
average country distance from the global centroid:
```

```
latest_gt %>% summarise(centroid_dist = round(mean(sqrt((gt1 - mean(gt1))^2 +  
                                                    (gt2 - mean(gt2))^2)), 3)) %>% print
```

```
# A tibble: 1 x 1
```

	centroid_dist
	<dbl>
1	0.399

```
#' ## Paired country bootstrap
```

```
#'
```

```
#' Resamples the forty countries with replacement, the same resampled set across  
# all models within each replicate. This captures uncertainty from generalizing  
# across countries, which dominates in this design; it does not separately  
# propagate the within-cell variability of the ten elicitations.
```

```
boot_level <- function(df, label) {  
  bd <- boot_paired(df, "dist")  
  report_ci(bd, sprintf("%s mean distance, 95%% CI (lower = closer)", label))  
  report_pairwise(bd, sprintf("%s accuracy", label))  
  bd  
}
```

```
bd_c1 <- boot_level(c1_latest, "C1")
```

=== C1 mean distance, 95% CI (lower = closer) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.202	0.238	0.278	ABOVE
claude-opus-4-8	0.146	0.177	0.208	ABOVE
gemini-3.6-flash	0.124	0.152	0.182	ABOVE
gpt-5.5	0.160	0.193	0.224	ABOVE

=== C1 accuracy: pairwise difference (row - col), 95% CI; * = excludes 0 ===

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	0.026	0.061	0.096	*
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.040	0.085	0.129	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	0.012	0.046	0.078	*
claude-opus-4-8	vs gemini-3.6-flash	-0.007	0.024	0.055	
claude-opus-4-8	vs gpt-5.5	-0.039	-0.016	0.006	
gemini-3.6-flash	vs gpt-5.5	-0.065	-0.040	-0.017	*

```
bd_c2 <- boot_level(c2_latest, "C2")
```

=== C2 mean distance, 95% CI (lower = closer) ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.199	0.231	0.266	ABOVE
claude-opus-4-8	0.150	0.182	0.215	ABOVE
gemini-3.6-flash	0.120	0.151	0.185	ABOVE
gpt-5.5	0.152	0.184	0.215	ABOVE

=== C2 accuracy: pairwise difference (row - col), 95% CI; * = excludes 0 ===

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	0.016	0.050	0.082	*
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.033	0.080	0.126	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	0.015	0.048	0.080	*
claude-opus-4-8	vs gemini-3.6-flash	-0.006	0.031	0.067	
claude-opus-4-8	vs gpt-5.5	-0.028	-0.002	0.023	
gemini-3.6-flash	vs gpt-5.5	-0.060	-0.032	-0.006	*

```
saveRDS(list(C1 = bd_c1, C2 = bd_c2, c1_latest = c1_latest, c2_latest = c2_latest),
  "level_bootstrap.rds")
```

```
#' ## Per-country placement error
```

```
#'
```

```
#' Accuracy varies more across countries than across models, so this is the more
```

```
#' informative cut. Averaged over the four models to describe the country rather
#' than any one system.

by_country <- c1_latest %>%
  group_by(country) %>% summarise(mean_err = round(mean(dist), 3), .groups = "drop") %>%
  arrange(mean_err)

cat("\n=== per-country mean placement error (C1), best and worst eight ===\n")
```

=== per-country mean placement error (C1), best and worst eight ===

```
print(head(by_country, 8)); print(tail(by_country, 8))
```

```
# A tibble: 8 x 2
  country mean_err
  <chr>      <dbl>
1 IRN      0.083
2 BRA      0.087
3 CAN      0.093
4 USA      0.095
5 KGZ      0.102
6 RUS      0.102
7 IRQ      0.104
8 PHL      0.119
```

```
# A tibble: 8 x 2
  country mean_err
  <chr>      <dbl>
1 MYS      0.272
2 CYP      0.273
3 COL      0.282
4 JPN      0.337
5 KOR      0.337
6 TWN      0.338
7 THA      0.35
8 EGY      0.352
```

```
#' Sensitivity: the pre-specified indicator-coverage rule would drop Iraq and
#' Hong Kong from the snapshot comparisons. Re-run without them to confirm the
#' model ranking and every significance verdict are unchanged.
```

```
c1_38 <- c1_latest %>% filter(!country %in% c("IRQ", "HKG"))
cat("\n=== sensitivity: C1 excluding IRQ and HKG (n =", n_distinct(c1_38$country), ") ===\n")
```

```
=== sensitivity: C1 excluding IRQ and HKG (n = 38 ) ===
```

```
bd_38 <- boot_paired(c1_38, "dist")
report_ci(bd_38, "C1 mean distance, 38 countries")
```

```
=== C1 mean distance, 38 countries ===
```

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.209	0.245	0.283	ABOVE
claude-opus-4-8	0.148	0.178	0.212	ABOVE
gemini-3.6-flash	0.127	0.156	0.188	ABOVE
gpt-5.5	0.165	0.196	0.229	ABOVE

```
report_pairwise(bd_38, "C1 accuracy, 38 countries")
```

```
=== C1 accuracy, 38 countries: pairwise difference (row - col), 95% CI; * = excludes 0 ===
```

		lo2.5	median	hi97.5	sig
Qwen/Qwen3.6-Plus	vs claude-opus-4-8	0.030	0.066	0.104	*
Qwen/Qwen3.6-Plus	vs gemini-3.6-flash	0.041	0.089	0.136	*
Qwen/Qwen3.6-Plus	vs gpt-5.5	0.015	0.050	0.083	*
claude-opus-4-8	vs gemini-3.6-flash	-0.011	0.024	0.054	
claude-opus-4-8	vs gpt-5.5	-0.040	-0.017	0.006	
gemini-3.6-flash	vs gpt-5.5	-0.067	-0.040	-0.014	*